

Dhruv Singh

+91 8084407155 — singh.dhruv9097@gmail.com — linkedin.com/in/dhruv-singh — github.com/dhruv9097

EDUCATION

Tribhuvan College, Nalanda University Centre <i>Bachelor of Science (Hons.) in Data Science</i> • GPA: 7.75 (10 scale) • Coursework: Machine Learning, Data Visualization, SQL, Python Programming	Neemrana, Rajasthan 2023 – Present
Symbiosis Public School <i>Senior Secondary (Class XII) PCM with Informatics Practices</i> • Percentage: 74%	Dhanbad, Jharkhand 2021 – 2023

EXPERIENCE & RESEARCH

OSCode <i>Web Development Intern</i> • Architected & Developed Full-Stack Features: Built responsive web interfaces using Next.js, React, and Tailwind CSS, and developed cross-platform mobile app screens using React Native (Expo). • Scalable Backend Engineering: Engineered robust REST APIs and microservices using NestJS and Node.js, leveraging Prisma ORM with PostgreSQL for efficient data management. • Performance & Background Processing: Implemented asynchronous background job processing and message queues using BullMQ and Redis to handle event registrations, notifications, and scalable mentor payouts. • Advanced Monorepo Management: Maintained a complex Turborepo monorepo environment with shared TypeScript packages (UI, schemas, analytics), improving code reusability and CI/CD build times across 9+ applications. • Search & AI Integrations: Integrated Typesense for lightning-fast mentor discoverability/search, and utilized the OpenAI SDK for advanced backend automations. • State Management & Validation: Ensured type safety and robust form handling across the stack using Zod, React Hook Form, Zustand, and TanStack React Query.	Noida, Uttar Pradesh 2026
Subharti University <i>Paper Presentation</i> • Authored and presented a research paper titled “A Modern Way to Generate Electricity: Floating Solar Farms”. • Analyzed the efficiency and environmental impact of floating solar photovoltaic systems compared to land-based alternatives.	Remote/Hybrid 2024
Codtech IT Solutions <i>Python Hackathon Participant</i> • Competed in a rigorous Python hackathon focusing on algorithmic efficiency and problem-solving. • Developed scripts to solve data processing challenges within strict time limits.	Remote 2024

PROJECTS

Social Media Automation Engine — <i>Python, Gemini Vision API, Pollinations.ai, BeautifulSoup</i> • Engineered a 7-phase multi-agent AI system to automate social media management spanning competitor intelligence, trend analysis, content creation, and AI image generation. • Built specialized agents: Spy Agent (7-day competitor scraping), Auditor Agent (gap analysis), Image Analyst (Gemini Vision API), Trend Hijack Agent (world trend scoring), Architect Agent (thread/reply drafts), Engagement Agent (Golden Hour protocol), and Image Generator (Pollinations.ai). • Architected a live/mock dual-mode pipeline with quota management, structured JSON reporting, and a Next.js social manager UI dashboard.

GitHub Trending Dashboard — *Python, BeautifulSoup, Pandas, Next.js, TypeScript*

- Built a full-stack dashboard that scrapes GitHub's daily trending page, computes a custom Hype Score (1.25x daily star growth), and surfaces insights in real time.
- Python backend uses BeautifulSoup and Pandas to parse, clean, and export repo data to CSV; Next.js/TypeScript frontend renders an interactive leaderboard.
- Designed a clean data pipeline from raw HTML scrape to a structured, auto-refreshing frontend view.

Physio-Detection System: Virtual Paint — *Python, OpenCV, MediaPipe*

- Developed a computer vision application allowing users to draw on-screen using hand gestures without physical contact.
- Leveraged MediaPipe for real-time hand-tracking and OpenCV for image processing and gesture rendering.
- Optimized gesture recognition algorithms to reduce latency and improve drawing precision.

Life Expectancy Analysis — *Python, Pandas, NumPy, Data Visualization*

- Conducted exploratory data analysis on a global life expectancy dataset to identify key health and socioeconomic indicators.
- Processed and cleaned multi-source datasets using Pandas and NumPy, handling missing values and outliers.
- Produced visualizations to reveal trends across countries and time periods, drawing actionable insights.

Personal Portfolio & Blog — *Next.js, TypeScript, Tailwind CSS, MDX*

- Built and deployed a personal portfolio and blog site using Next.js (App Router) and TypeScript with Tailwind CSS styling.
- Integrated MDX for dynamic blog post rendering; deployed on Vercel at sorted-blog.vercel.app.

Academic Case Studies — *Research, Data Analysis*

- Microsoft Farmbeats: Analyzed the integration of IoT and AI in precision agriculture to improve crop yields.
- Waste to Wealth: Investigated bioplastic production methodologies as sustainable alternatives to synthetic plastics.

TECHNICAL SKILLS

- **Languages:** TypeScript, Python (Intermediate), C++ (Basic), SQL (Basic)
- **Frontend & Mobile:** Next.js, React, React Native (Expo), Tailwind CSS
- **Backend & Databases:** Node.js, NestJS, PostgreSQL, Prisma ORM, Redis
- **Libraries & State Management:** Zustand, TanStack React Query, React Hook Form, Zod, OpenCV, MediaPipe, Pandas, NumPy, BeautifulSoup
- **Architecture & Tools:** Turborepo, BullMQ, Typesense, QGIS, Git, GitHub
- **AI & APIs:** Gemini Vision API, Pollinations.ai, OpenAI SDK, Multi-Agent Systems, Prompt Engineering
- **Domains:** Full-Stack Web/Mobile Development, Data Analysis, Data Visualization, Machine Learning, Computer Vision, AI Automation

LEADERSHIP & EXTRACURRICULAR

Tribhuvan College

Neemrana, Rajasthan

Event Organizer

2023

- Coordinated the “Principal’s Meet,” a summit discussing “Changing Climate & Environment” for local school principals.
- Organized an inter-school debate competition, managing logistics and participant coordination.